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Mediation, Moderation, and Conditional Process Analysis

A Comprehensive Practice Worksheet

How to Use This Worksheet

This worksheet covers major concepts in mediation and moderation analysis. Each module includes:

- **The core equations** with plain-English translations
- **Worked numerical examples** from PROCESS macro output (see lecture notes)
- **Fill-in-the-blank problems** to test yourself
- **Interpretation prompts** to practice writing results

Work through the modules in order. Later modules build on earlier ones.

Module 1: Simple Mediation — The Foundation

What Is Mediation?

Mediation answers the question: **Why does X affect Y?** It proposes that X doesn't just directly cause Y — it does so *through* an intermediate variable (the **mediator**, M).

Case Study: Violence Exposure and Child Externalizing Behavior (NASCW Data). The research questions are whether: (1) children who have **witnessed severe violence at Wave 1 (EV_W1)** are more likely to engage in **externalizing behaviour at Wave 3 (bcext3)** ; and (2) **post-traumatic stress at Wave 1 (tra1)** is the mechanism through which violence exposure is linked to externalizing behaviour.

Classic example: Violence exposure (X) → Traumatic stress (M) → Behavioral problems (Y)

The Two Core Equations

Equation 1 — Predicting the Mediator:

$$M = i_M + aX + e_M$$

In plain English: A child's traumatic stress (M) is predicted by their violence exposure (X). The coefficient *a* tells us how much trauma increases for each unit increase in violence exposure.

Equation 2 — Predicting the Outcome:

$$Y = i_Y + c'X + bM + e_Y$$

In plain English: Behavioral problems (Y) are predicted by *both* violence exposure (X) and traumatic stress (M) simultaneously. The coefficient *c'* (pronounced "c prime") is the **direct effect** of X on Y after accounting for M. The coefficient *b* captures how much behavior problems increase per unit of trauma.

The Three Key Paths

Symbol	Path	Meaning
<i>a</i>	X → M	How much does X increase M?
<i>b</i>	M → Y (controlling for X)	How much does M increase Y?
<i>c'</i>	X → Y (controlling for M)	Direct effect of X

Symbol	Path	Meaning
ab	$X \rightarrow M \rightarrow Y$	Indirect effect (mediation)
c	$X \rightarrow Y$ (total)	Total effect = $c' + ab$

Worked Example (from Slide 6 Output)

Given values:

- Path a (Violence $\rightarrow$ Trauma): **1.9423**, $p < .001$
- Path b (Trauma $\rightarrow$ Behavior, controlling for Violence): **0.0976**
- Direct effect c' (Violence $\rightarrow$ Behavior, controlling for Trauma): **0.8237**

Problem 1.1 — Calculate the Indirect Effect (ab):

$$a \times b = 1.9423 \times 0.0976 = \underline{\hspace{2cm}}$$

✓ *Check:* You should get approximately **0.1896**

Problem 1.2 — Calculate the Total Effect (c):

$$c = c' + ab = 0.8237 + 0.1896 = \underline{\hspace{2cm}}$$

✓ *Check:* This should match the regression coefficient from a model that **excludes** the mediator (i.e., regressing Y on X alone).

Concept Check

Q: What would it mean if the direct effect c' became non-significant after adding M to the model, but the indirect effect ab was significant?

This is called **full mediation** — the relationship between X and Y is *entirely* explained by M . If c' remains significant alongside a significant ab , this is **partial mediation**.

Here is a write-up of the [results](#).

Module 2: Mediation with Covariates

Why Include Covariates?

Without controlling for background variables (e.g., gender, experience, self-efficacy), you risk **confounding**: the mediation effect you observe might just be a spurious byproduct of some third variable that was never measured.

Case Study: Economic Stress Among Small Business Owners. Older business owners might have both more stress *and* more tenure. If you don't control for tenure, you can't be sure whether the stress effect is real or just a proxy for experience. Participants were N=262 entrepreneurs who were members of **Business Networking International**, who responded to an online survey about their business performance and emotional reactions to the economic climate. Pollack and Hayes proposed that economic stress leads to depressed affect, which in turn enhances a business owner's intention to leave entrepreneurship. Does ESTRESS affect withdrawal intentions *through* depressed affect? Is AFFECT the mechanism?

The Statistical Model (Model 4 with Covariates)

Note: The general equations are shown on slide 18.

Equation for the Mediator (M):

$$M = i_M + aX + \gamma_1 ESE + \gamma_2 Sex + \gamma_3 Tenure + e_M$$

Equation for the Outcome (Y):

$$Y = i_Y + c'X + bM + h_1 ESE + h_2 Sex + h_3 Tenure + e_Y$$

In plain English: We statistically "partial out" the influence of gender, self-efficacy, and tenure from both M and Y. If the *a* and *b* paths remain significant after this adjustment, we have a much stronger case for mediation.

 **Common Misconception:** Covariates appear in the equations above but are **not** part of the indirect effect calculation. The indirect effect is always $a \times b$ — the

covariates simply help produce more accurate estimates of a and b .

Practice Problems

Problem 2.1 — Interpretation (Slide 18 Output):

The coefficient for ESE (self-efficacy) on withdrawal (Y) is negative and significant ($p < .05$).

Q: In plain language, what does a *negative* ESE coefficient mean for withdrawal behavior?

Answer: Higher self-efficacy is associated with *lower* withdrawal. In other words, owners who believe in their own competence are less likely to disengage from their business.

Problem 2.2 — Identifying Variables (Slide 17 for SPSS Setup):

Role	Variable
X (Predictor)	_____
Y (Outcome)	_____
Covariates	_____

Problem 2.3 — Derivation:

Given: $a = 0.45$, $b = -0.30$, coefficient for Gender on M = 0.15

- What is the **indirect effect**? _____
- Does the Gender coefficient change the value of ab ? [YES / NO]

Hint: Covariates improve the *accuracy* of a and b estimates, but they are not multiplied into the $a \times b$ calculation itself.

Problem 2.4 — Output Analysis (Slide 20, Outcome: Affect):

- Coefficient for ESE on affect: _____
- Is it significant ($p < .05$)? _____

- If ESE reduces affect (the mediator), what happens to withdrawal intention (Y)? Explain.

Module 3: Parallel and Serial Mediation

3A. Parallel Mediation (Model 4)

In parallel mediation, X influences Y through **two or more mediators that operate independently** — they don't communicate with each other.

The Equations:

$$M_1 = i_{M_1} + a_1X + e_{M_1}$$

$$M_2 = i_{M_2} + a_2X + e_{M_2}$$

$$Y = i_Y + c'X + b_1M_1 + b_2M_2 + e_Y$$

Indirect Effects:

Path	Calculation	Meaning
Via M_1	$a_1 \times b_1$	X affects Y through M_1
Via M_2	$a_2 \times b_2$	X affects Y through M_2
Total Indirect	$a_1b_1 + a_2b_2$	Combined mediated effect

Case Study: Presumed Media Influence (PMI). N=123 Israelis (43 male, 80 female) read an article about a potential sugar shortage. Half (n=58) were told it would appear on the **FRONT PAGE** of a major newspaper; half (n=65) were told it was in the **MIDDLE OF AN ECONOMIC SUPPLEMENT**. Same article — only described placement differed. Participants reported their intention to buy sugar (REACTION) and their belief that others would be influenced by the media (PMI). Front-page placement doesn't directly drive buying. It works through two psychological mechanisms: importance (IMPORT) and presumed media influence (PMI).

Worked Example (Slide 36):

- a_1b_1 (via Perceived Importance): $0.4765 \times 0.3965 = \mathbf{0.1896}$
- a_2b_2 (via Perceived Social Influence): $0.3244 \times 0.6268 = \mathbf{0.2303}$
- **Total Indirect Effect:** $0.1896 + 0.203 = \mathbf{0.392}$

In plain English: A news article's front-page placement influences purchase intention through two *separate* and *simultaneous* channels — readers think the issue matters personally (M_1), AND they think their friends will be influenced (M_2). These two mechanisms don't depend on each other.

3B. Serial Mediation (Model 6)

In serial mediation, the mediators influence **each other in a specific sequence** — creating a chain or "domino effect."

The Equations:

$$M_1 = i_{M_1} + a_1X + e_{M_1}$$

$$M_2 = i_{M_2} + a_2X + d_{21}M_1 + e_{M_2}$$

$$Y = i_Y + c'X + b_1M_1 + b_2M_2 + e_Y$$

Note the key difference: M_2 is now predicted by *both* X and M_1 . The coefficient d_{21} captures how strongly M_1 drives M_2 .

Three Indirect Effects in Model 6:

Path	Calculation	Description
Indirect 1	$a_1 \times b_1$	$X \rightarrow M_1 \rightarrow Y$
Indirect 2	$a_2 \times b_2$	$X \rightarrow M_2 \rightarrow Y$
Indirect 3 (serial)	$a_1 \times d_{21} \times b_2$	$X \rightarrow M_1 \rightarrow M_2 \rightarrow Y$

Worked Example (Slide 45):

- Indirect 1: $___ \times ___ = \mathbf{0.203}$
- Indirect 2: $___ \times ___ = \mathbf{0.141}$
- Indirect 3 (serial chain): $___ \times ___ \times ___ = \mathbf{0.049}$

In plain English: Seeing the article on the front page (X) makes readers feel the news is *important* (M₁). That sense of importance then increases belief that *others* will be swayed (M₂), which finally drives purchase intention (Y). The critical distinction from parallel mediation: M₁ causes M₂.

Module 4: Simple Moderation

What Is Moderation?

Moderation answers the question: **For whom** (or **when**) does X affect Y? A moderator (W) changes the *strength or direction* of the X → Y relationship.

Case Study: Sexual Minority Status and Delinquency. We know that there is an association between PTSS and engaging in delinquent behavior, but is the effect of PTSS on delinquent behavior **stronger for sexual minority youth**? This question is explored below.

The Statistical Model

$$Y = i_Y + b_1X + b_2W + b_3XW + e_Y$$

The term XW is the **interaction** — it captures how the combination of X and W together predicts Y, above and beyond either variable alone. If b_3 is significant, the X → Y effect depends on W.

Interpreting the Results (Slide 61)

Group	Effect of Trauma on Delinquency	Significance
Heterosexual Youth	-0.0450	$p = .4673$ (non-significant)
Sexual Minority Youth	+0.2130	$p = .0514$ (significant)

Interaction term $b_3 = 0.2580, p < .05$

In plain English: Trauma has a meaningful impact on delinquency *only* for sexual minority youth. For heterosexual youth, the same level of trauma shows no statistically significant link to delinquency. The interaction term confirms this difference is not due to chance.

Module 5: Johnson-Neyman (Region of Significance)

What Problem Does J-N Solve?

The "pick-a-point" approach (Module 8) tests the $X \rightarrow Y$ relationship at only three arbitrary values of W (e.g., -1 sd below the mean, the mean, +1 sd above the mean OR the 16th, 50th and 84th percentiles). The **Johnson-Neyman technique** instead finds the *exact values of W* at which the effect transitions between significant and non-significant.

Key Output (Slide 68)

- **Transition point (cut-off): $W = 8.43$**
- The effect of trauma on delinquency is significant when age > 8.43

Practice Questions:

Q: Does trauma significantly predict delinquency for a 10-year-old child? [**YES / NO**] (*Hint: $10 > 8.43$*)

Q: Does trauma significantly predict delinquency for a 5-year-old child? [**YES / NO**]

Module 6: Moderated Mediation — Model 7

What Is Moderated Mediation?

Moderated mediation (sometimes called **conditional process analysis**) combines both mediation and moderation. In **Model 7**, the moderator W affects the a path — meaning how strongly X predicts M depends on W .

The Equations

$$M = i_M + a_1X + a_2W + a_3XW + e_M$$

$$Y = i_Y + c'X + bM + e_Y$$

In plain English: A child's violence exposure (X) leads to trauma (M), which predicts behavior problems (Y). But the *strength* of the violence → trauma link (a_1) differs across groups defined by W (e.g., gender or age). The interaction term a_3 tests this.

Numerical Example (Slide 64)

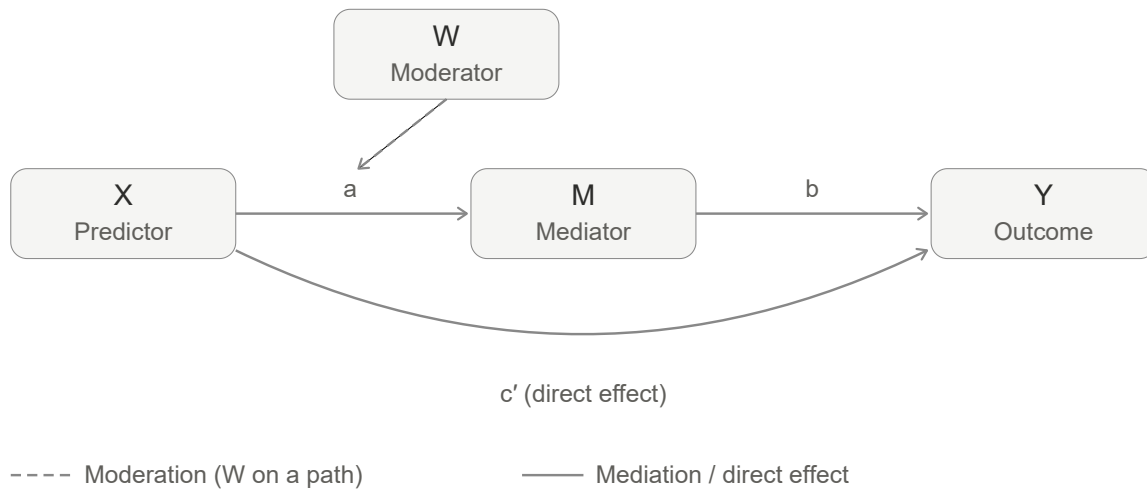
Given: $a_1 = 0.40$ (overall violence → trauma effect), $a_3 = 0.15$ ($p < .05$)

Group	a Path Calculation	Resulting Effect
Group 1 ($W = 0$)	$0.40 + (0.15 \times 0)$	0.40
Group 2 ($W = 1$)	$0.40 + (0.15 \times 1)$	0.55

Interpretation: The violence → trauma link is **37.5% stronger** for Group 2 than Group 1. Consequently, the *indirect effect* ab also differs across groups — this is the definition of moderated mediation.

 **Key insight:** In Model 7, you should report the **conditional indirect effects** (the ab product at specific values of W), not just the a_3 coefficient alone.

Conceptual Diagram (Model 7)



Module 7: Comparing Models 4, 7, and 8

Model	What Is Moderated?	Key Question
Model 4	Nothing — simple mediation	Does M explain why $X \rightarrow Y$?
Model 7	The a path ($X \rightarrow M$)	Does W change <i>how strongly</i> X predicts M?
Model 8	Both a path AND c' path	Does W change both the mechanism <i>and</i> the direct effect?

Model 8 Equations (Slides 69–70)

$$M = i_M + a_1X + a_2W + a_3XW + e_M$$

$$Y = i_Y + c'_1X + c'_2W + c'_3XW + bM + e_Y$$

In plain English: Model 8 is "moderation everywhere." Age (W) changes both how much trauma children develop from violence exposure *and* how strongly violence directly affects behavior. Use this model when theory predicts W operates at multiple stages.

Module 8: Probing Interactions — Pick-a-Point

When J-N is unavailable or the data don't support it, researchers "pick points" — typically the **16th percentile (Low)**, **Mean**, and **84th percentile (High)** of the moderator — and test the $X \rightarrow Y$ simple slope at each.

Example output:

Level of W	Effect of X on Y	p-value	Significant?
Low (16th %ile)	0.10	.15	No
Mean	0.30	.03	Yes
High (84th %ile)	0.50	.001	Yes

Critical Analysis Question:

If you only reported the "High" group result, would you be giving readers a complete picture of the data?

No. Reporting only the significant subgroup is selective and misleading. Readers would not know that the effect is absent at low levels of W — which is theoretically and practically important.

Module 9: Assumptions to Check Before Running PROCESS

Before running any mediation or moderation analysis, verify these assumptions on your data:

Assumption	What to Check	What Goes Wrong If Violated
Linearity	Scatterplots, residual plots	Pathsa, b , c' are biased estimates
Normality of residuals	Q-Q plots, Shapiro-Wilk test	Standard errors are unreliable

Assumption	What to Check	What Goes Wrong If Violated
Homoscedasticity	Residual vs. fitted plots	Confidence intervals are inaccurate
No multicollinearity	VIF values	Estimates become unstable

Practical note: Bootstrap confidence intervals (used by PROCESS by default) are robust to some violations — but they don't fix everything. Always examine your data first.

Module 10: Power and Sample Size

The indirect effect (ab) is harder to detect than a simple correlation — because you are estimating a *product* of two regression coefficients, each with its own sampling variability.

Model Complexity	Recommended Minimum N
Simple mediation (Model 4)	N > 100
Parallel mediation (2 mediators)	N > 200
Serial or moderated mediation (Models 6, 7, 8)	N > 300

Significance Decision Rule:

A bootstrap confidence interval for ab is significant if and only if **zero is not contained** in the interval.

Example: $ab = 0.05$, 95% BCa CI $[-0.01, 0.12]$ → **NOT significant** (zero falls inside the interval)

Module 11: Research Design Exercise

Design your own mediation-moderation study on the following topic:

Research question: How does social media usage (X) lead to poor sleep (Y)?

Element	Your Answer
Mediator (M)	(e.g., anxiety, rumination, blue light exposure)
Moderator (W)	(e.g., age, screen time awareness, sleep hygiene habits)
Hayes PROCESS Model	Model _____

Write the equation for Y:

$Y =$ _____

Part 2: Reporting Templates

Use these templates when writing up results. Fill in the blanks from your PROCESS output.

Template 1 — Parallel Mediation (Model 4)

☒ influenced [Y] through [M₁]: $ab =$ _____, 95% BCa CI [_____, _____] ☒ also influenced [Y] through [M₂]: $ab =$ _____, 95% BCa CI [_____, _____]. Because neither confidence interval contains zero, both indirect pathways are statistically significant."

Template 2 — Serial Mediation (Model 6)

☒ predicted [M₁] ($a_1 =$ _____), [M₁] in turn predicted [M₂] ($d_{21} =$ _____), and [M₂] predicted [Y] ($b_2 =$ _____). The specific indirect effect through the full serial chain ($a_1 \times d_{21} \times b_2$) was _____, 95% BCa CI [_____, _____], indicating a statistically significant domino-style mechanism."

Template 3 — Simple Moderation (Model 1)

✓ on [Y] varied by [W]. The interaction term was significant ($b = \underline{\hspace{1cm}}$, $p = \underline{\hspace{1cm}}$). Simple slopes analysis revealed that the [X]–[Y] relationship was significant for [Group A] ($b = \underline{\hspace{1cm}}$, $p = \underline{\hspace{1cm}}$) but not for [Group B] ($b = \underline{\hspace{1cm}}$, $p = \underline{\hspace{1cm}}$), suggesting the effect is conditional on [W]."

Part 3: Advanced Scenario — Mental Health Case Study

Variables:

- X = Childhood adversity
- M = Emotional dysregulation
- W = Social support
- Y = Adult quality of life

Question 3.1:

If a_3 (the XW interaction on M) is significant in a Model 7 analysis, what does that tell you about social support?

It means that social support **moderates the pathway** from adversity to emotional dysregulation. High social support may buffer children from developing dysregulation even when adversity is high — the *a* path is weaker for those with more support. You should report conditional indirect effects at different levels of W.

Question 3.2:

If the J-N technique shows the effect is only significant when social support is **below 15**, what would you recommend clinically?

Intervention efforts should prioritize individuals with low social support (scores below 15). Above that threshold, the adversity → dysregulation → quality of life chain is no longer statistically meaningful — suggesting that adequate social support naturally disrupts the mechanism without additional intervention.

Part 4: Final Synthesis

4.1 — The Big Picture

Concept	Core Question	Answers "Why" or "When"?
Mediation	How does X get to Y?	WHY X affects Y
Moderation	For whom / under what conditions?	WHEN (or for whom) X affects Y

4.2 — Case Study Matching

Match each case study to the correct PROCESS Model:

#	Case Study Description	Model
1	Violence Exposure → Trauma → Behavior	Model _____
2	Article Front Page → Importance → Influenced Others → Purchase	Model _____
3	Stress → Affect → Withdrawal (controlling for Gender, ESE, Tenure)	Model _____
4	Trauma → Delinquency, moderated by Sexual Minority Status	Model _____

(Answers: 4, 6, 4 with covariates, 1)

4.3 — PROCESS Variable Entry (Model 6 — PMI Study)

PROCESS Field	Variable Name
Y (Outcome)	_____
X (Predictor)	_____
M ₁ (First Mediator)	_____
M ₂ (Second Mediator)	_____

4.4 — Interpretation

Given: Path $a = 1.5$, Path $b = 0.10$, Direct path $c' = 0.80$, Interaction term XW is **non-significant**

Q1: Why is this **simple mediation** and NOT moderated mediation?

Because the interaction term XW is non-significant, there is no evidence that the $X \rightarrow M$ or $X \rightarrow Y$ relationships differ across levels of W . Without a significant interaction, moderation cannot be claimed.

Q2: How many indirect effects are there and why?

There is **one** indirect effect: $X \rightarrow M \rightarrow Y$, calculated as $a \times b = 1.5 \times 0.10 = \mathbf{0.15}$. Simple mediation with one mediator produces exactly one indirect pathway.

Q3: Name the paths specifically:

- a path: $X \rightarrow M$ (effect = 1.5)
- b path: $M \rightarrow Y$, controlling for X (effect = 0.10)
- c' path: $X \rightarrow Y$, controlling for M (direct effect = 0.80)
- ab (indirect effect) = 0.15
- c (total effect) = $0.80 + 0.15 = \mathbf{0.95}$

Appendix: Manual Calculation Practice Lab

Scenario

A researcher finds the following path coefficients:

- Violence exposure $\rightarrow$ Trauma (a) = **0.50**
- Trauma $\rightarrow$ Behavior problems (b) = **0.30**
- Violence exposure $\rightarrow$ Behavior (direct, c') = **0.10**

Problem A.1 — Indirect Effect:

$ab = 0.50 \times 0.30 = \underline{\hspace{2cm}}$

Problem A.2 — Total Effect:

$c = c' + ab = 0.10 + \underline{\hspace{1cm}} = \underline{\hspace{2cm}}$

Problem A.3 — Proportion Mediated:

Proportion = $\frac{ab}{c} = \frac{\underline{\hspace{1cm}}}{\underline{\hspace{1cm}}} = \underline{\hspace{2cm}}$

This tells you what percentage of the *total* effect of violence on behavior problems is *explained by* the trauma pathway.

Answer Key

Problem	Answer	Notes
A.1	0.15	$ab = 0.50 \times 0.30$
A.2	0.25	$c = 0.10 + 0.15$
A.3	60%	$0.15 / 0.25 = 0.60$
Module 1.1	0.1896	1.9423×0.0976
Module 1.2	1.0133	$0.8237 + 0.1896$
Module 2.3	$ab = -0.135$	0.45×-0.30 ; Gender does NOT change ab
J-N age 10	YES	$10 > 8.43$ transition point
J-N age 5	NO	$5 < 8.43$ transition point
CI [-0.01, 0.12]	NOT significant	Zero is inside the interval

End of Worksheet